

Asymptotic Results

STAT 24510-30040

(Winter 2025)

Standard theoretical results on convergence properties of random variables are frequently applied in statistics in order to obtain useful results for large samples, such as constructing asymptotic confidence intervals.

This note states common theoretical asymptotic results without proofs. The assumptions stated are common conditions, not necessarily the weakest ones.

First, we describe the types of convergence of a sequence of random variables.

Let $\{X_1, X_2, \dots\} = \{X_n\}_{n=1}^\infty$ be a sequence of random variables with cumulative probability distribution functions

$$F_n(x) = \mathbb{P}(X_n \leq x).$$

Let X be a random variable with cumulative distribution function

$$F(x) = \mathbb{P}(X \leq x).$$

Convergence in distribution

The sequence $\{X_1, X_2, \dots\}$ is said to converge in distribution to X if

$$\lim_{n \rightarrow \infty} F_n(x) = F(x)$$

at any continuous point x of $F(x)$.

Notations

Convergence in distribution can be stated as

$$X_n \rightarrow X \quad \text{in distribution as } n \rightarrow \infty.$$

Common notations are

$$X_n \xrightarrow{\mathcal{D}} X \quad \text{as } n \rightarrow \infty$$

or

$$X_n \xrightarrow[n \rightarrow \infty]{\mathcal{D}} X.$$

Remarks on convergence in distribution

- Recall that the cumulative distribution functions (CDF) of a random variable $F_X(x) = \mathbb{P}(X \leq x)$ is a non-decreasing function defined on $(-\infty, \infty) = \mathbb{R}$, and the range or possible values of $F_X(x)$ is $[0, 1]$.

Cumulative distribution functions of real valued continuous random variables are continuous on $\mathbb{R} = (-\infty, \infty)$:

$$\lim_{x \rightarrow x-} F_X(x) = \lim_{x \rightarrow x+} F_X(x) = F_X(x), \quad \forall x \in \mathbb{R}.$$

For real valued discrete random variables, cumulative distribution functions are right-continuous step functions:

$$\lim_{x \rightarrow x-} F_X(x) \leq \lim_{x \rightarrow x+} F_X(x) = F_X(x), \quad \forall x \in \mathbb{R}.$$

The step functions is right-continuous only at points with non-zero probability and continuous elsewhere:

$$P(X = x_0) > 0 \quad \iff \quad \lim_{x \rightarrow x-} F_X(x) < \lim_{x \rightarrow x+} F_X(x) = F_X(x)$$

The requirement of the convergence only at continuous points makes the results of convergence-in-distribution theorems applicable to distributions of discrete random variables as well as continuous ones.

- Other names for Convergence-in-Distribution are Convergence-in-Law, or Convergence-Weakly.
- Convergence in distribution does not necessarily imply convergence of the corresponding probability density functions.

Convergence in Probability (to a constant)

If a sequence of random variables converges in distribution to a constant, i.e., a degenerated random variable,

$$X_n \xrightarrow[n \rightarrow \infty]{\mathcal{D}} X, \quad X \equiv a$$

then the sequence is said to converge to the constant a in probability, written as

$$X_n \xrightarrow[n \rightarrow \infty]{\mathcal{P}} a$$

Remarks on convergence in probability

- The above shows that, for constant limit a ,

$$X_n \xrightarrow[n \rightarrow \infty]{\mathcal{P}} a \quad \iff \quad X_n \xrightarrow[n \rightarrow \infty]{\mathcal{D}} a$$

- Convergence in probability to a constant can also be stated as

$$\lim_{n \rightarrow \infty} \mathbb{P}(|X_n - a| > \varepsilon) = 0, \quad \forall \varepsilon > 0.$$

Or equivalently,

$$\lim_{n \rightarrow \infty} \mathbb{P}(|X_n - a| < \varepsilon) = 1, \quad \forall \varepsilon > 0.$$

- Here we focus on convergence in probability to a constant only. General convergence to a non-degenerating random variable is a type of convergence with substantial implications on the joint distributions (omitted in this course).

Convergence With Probability One

The sequence $\{X_1, X_2, \dots\}$ is said to converge to X with probability one if

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} X_n = X\right) = 1$$

and can be written as

$$X_n \xrightarrow[n \rightarrow \infty]{w.p.1} X$$

or equivalently

$$X_n \xrightarrow[n \rightarrow \infty]{a.s.} X$$

where *a.s.* stands for *almost surely*.

Remarks on convergence with probability one

- Convergence with probability 1 is also called convergence almost surely or almost everywhere.
- The definition can be more clearly defined in terms of the elements in the probability space Ω ,

$$\mathbb{P}\left(\omega \in \Omega : \lim_{n \rightarrow \infty} X_n(\omega) = X(\omega)\right) = 1$$

The Law of Large Numbers (LLN)

If $X_1, X_2, \dots$ are independent identically distributed random variables with mean μ and variance σ , then

$$\bar{X}_n \xrightarrow[n \rightarrow \infty]{\mathcal{P}} \mu$$

where $\bar{X}_n = \sum_{i=1}^n X_i/n$ is the sample mean.

Remarks on the Law of Large Numbers

- The law of large numbers states that with a sufficiently large sample, the average of the observations will be close to the expected value with probability very close to 1.

$$\lim_{n \rightarrow \infty} \mathbb{P}(|\bar{X}_n - \mu| < \varepsilon) = 1, \quad \forall \varepsilon > 0.$$

- The stated version is called the weak law of large numbers, which is sufficient for results used in this course. The condition for the weak law of strong numbers to hold can be weaker (omitted).
- The Strong Law of Large Number (SLLN) states the stronger version of the law of large numbers, in terms of almost sure convergence:

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} X_n = \mu\right) = 1$$

under certain conditions on second moments (omitted).

The Central Limit Theorem

Assume $X_1, X_2, \dots$ are independent identically distributed random variables with mean μ and variance σ^2 .

Denote $\bar{X}_n = \sum_{i=1}^n X_i/n$, then

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow[n \rightarrow \infty]{\mathcal{D}} Z, \quad Z \sim N(0, \sigma^2)$$

Equivalently,

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow[n \rightarrow \infty]{\mathcal{D}} Z, \quad Z \sim N(0, 1)$$

Remarks on the Central Limit Theorem

- The Central Limit Theorem says, as n becomes sufficiently large, $\sqrt{n}(\bar{X}_n - \mu)/\sigma$ is approximately of standard normal distribution, which can be stated as

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \sim N(0, 1) \quad \text{asymptotically.}$$

- Consequently, for large n , $\bar{X}_n$ is approximately of normal distribution with mean μ and variance σ^2/n . This is often loosely stated as

$$\bar{X}_n \sim N\left(\mu, \frac{\sigma^2}{n}\right) \quad \text{for large } n.$$

- By the central limit theorem, the mean of i.i.d. sample of finite variance is approximately of normal distribution as long as the sample size is sufficiently large, even when the original random variables themselves are not normally distributed.

Therefore many normal distribution based theoretical results are applicable to problems related to other distributions.

- There are many random variables not having finite variance, or even finite mean.

The most famous example is the Cauchy random variable. It has a well defined density function, e.g.,

$$f(x) = \frac{1}{\pi(1+x^2)}$$

but not finite mean (let alone finite variance).

Slutsky's Theorem

Assume a sequence of random variables

$$X_n \xrightarrow[n \rightarrow \infty]{\mathcal{D}} X$$

and another sequence of random variables

$$Y_n \xrightarrow[n \rightarrow \infty]{\mathcal{P}} a$$

convergent to a constant $a \in \mathbb{R}$ in probability.

If $g(x, y)$ is a function jointly continuous at (x, a) for all $x \in \mathbb{R}$, then

$$g(X_n, Y_n) \xrightarrow[n \rightarrow \infty]{\mathcal{D}} g(X, a)$$

Corollaries of Slutsky's Theorem

Assume

$$X_n \xrightarrow{\mathcal{D}} X, \quad Y_n \xrightarrow{\mathcal{P}} a \quad \text{as } n \rightarrow \infty,$$

Then

$$X_n + Y_n \xrightarrow[n \rightarrow \infty]{\mathcal{D}} X + a$$

$$X_n Y_n \xrightarrow[n \rightarrow \infty]{\mathcal{D}} aX$$

and if $a \neq 0$,

$$X_n/Y_n \xrightarrow[n \rightarrow \infty]{\mathcal{D}} X/a$$

Remarks on Slutsky's Theorem and its corollaries

- Slutsky's Theorem and its corollaries provide desirable limiting behavior for functions of two random sequences, as long as one sequence converges to a constant.
- If none of the two limits are constants, convergence properties heavily depend on joint distribution assumptions thus much harder to achieve, even for simple functions such as sums.
- Slutsky's theorem can be extended to multiple sequences.

Examples of asymptotic approximations

The normal approximation to sums of random variables stated in the Central Limit Theorem is applicable to discrete random variables as well as the continuous ones.

In the following we look into the normal approximation of binomial random variables.

Binomial by normal approximation

Let $Y \sim \text{Bin}(n, p)$, the binomial distribution with parameters (n, p) . Y can be viewed as the sum of n i.i.d. Bernoulli random variables $X_1, \dots, X_n$ with parameter p .

By the Central Limit Theorem,

$$\bar{X}_n \sim N\left(\mu, \frac{\sigma^2}{n}\right) \quad \text{for large } n,$$

with $\mu = p, \sigma^2 = p(1-p)$.

By the definition of binomial, $Y = n\bar{X}$ with mean np and variance $np(1-p)$, hence

$$Y \sim N(np, np(1-p)) \quad \text{for large } n.$$

Equivalently,

$$\frac{Y - np}{\sqrt{np(1-p)}} \sim N(0, 1) \quad \text{for large } n.$$

Using common notations for the standard normal random variable, its density and cumulative distribution functions,

$$Z \sim (0, 1), \quad \Phi(z) = \mathbb{P}(Z \leq z) = \int_{-\infty}^z \phi(t) dt = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

the cumulative probability function of binomial Y can be approximated by that of the standard normal.

$$\mathbb{P}(Y \leq k) = \mathbb{P}\left(\frac{Y - np}{\sqrt{np(1-p)}} \leq \frac{k - np}{\sqrt{np(1-p)}}\right) \approx \mathbb{P}\left(Z \leq \frac{k - np}{\sqrt{np(1-p)}}\right) = \Phi\left(\frac{k - np}{\sqrt{np(1-p)}}\right) \quad (1)$$

For a continuous random variable, the probability at one single point is zero, since integral of a continuous density function over an interval of length zero is zero.

To approximate the probability of a discrete random variable at a point by a continuous random variable, we choose to view the point mass probability as the probability of the discrete random variable over an appropriate interval.

For $Y \sim \text{Bin}(n, p)$, since Y only has non-zero probability at integer values, we may write

$$\mathbb{P}(Y = k) = \mathbb{P}\left(Y \in (k-1, k]\right) = \mathbb{P}(k-1 < Y \leq k)$$

The left-open, right-close interval format allows the total probability to be expressed as a union of disjoint intervals covering the whole real line $\mathbb{R}$. Then Y is well defined on $\mathbb{R}$, just as a continuous random variable is well defined at every point in $\mathbb{R}$.

The probability of Y over a left-open, right-continuous interval can be written as the difference of its cumulative distribution function at the end points of the interval,

$$\mathbb{P}(k-1 < Y \leq k) = \mathbb{P}(Y \leq k) - \mathbb{P}(Y \leq k-1)$$

which can be approximated by the cumulative distribution function $\Phi(z)$ of the standard normal, as derived in (1),

$$\mathbb{P}(Y \leq k) \approx \Phi\left(\frac{k - np}{\sqrt{np(1-p)}}\right)$$

By representing point mass probability as the probability over an interval with integer endpoints, the probability mass function of discrete binomial random variable Y can be approximated (without further continuity correction) by normal random variable as

$$\mathbb{P}(Y = k) = \mathbb{P}(Y \leq k) - \mathbb{P}(Y \leq k-1) \approx \Phi\left(\frac{k - np}{\sqrt{np(1-p)}}\right) - \Phi\left(\frac{k-1 - np}{\sqrt{np(1-p)}}\right)$$

Continuity correction in CLT normal approximation (Binomial example)

The above normal approximation of binomial is intuitive and simple. However due to the discreteness, the approximated probability by normal can be off, especially when the sample size n is moderate, not very large.

Note that we may also write the discrete probability as in an length-one interval centered at the discrete integer value:

$$\mathbb{P}(Y = k) = \mathbb{P}\left(Y \in \left(k - \frac{1}{2}, k + \frac{1}{2}\right]\right)$$

It turns out that this shift leads to a better approximation, called approximation with **continuity correction**.

In the binomial example, the normal approximation with continuity correction yields

$$\mathbb{P}(Y = k) = \mathbb{P}\left(Y \leq k + \frac{1}{2}\right) - \mathbb{P}\left(Y \leq k - \frac{1}{2}\right) \approx \Phi\left(\frac{k + \frac{1}{2} - np}{\sqrt{np(1-p)}}\right) - \Phi\left(\frac{k - \frac{1}{2} - np}{\sqrt{np(1-p)}}\right)$$

When applying CLT to discrete random variables, almost always, the CLT version with continuity correction is a better approximation than the one without continuity correction.

Reference: Section 5.5 in Casella & Berger; sections 5.1-5.4, 8.5 in Rice.